

Chenchen Mao

Ph.D. Candidate in Computer Science with expertise in human-AI interaction, social computing, behavioral experimentation, and quantitative analysis. Builds experimental systems and user-facing prototypes to study trust, perception, and behavior in AI-mediated and online platform environments.

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EDUCATION

Lehigh University, Bethlehem, PA

Ph.D. in Computer Science | Expected Dec 2026

- Ph.D. Candidate; all coursework and candidacy requirements completed.

Syracuse University, Syracuse, NY

M.S. in Computer Science | May 2021

University of Denver, Denver, CO

B.S. in Computer Science | Dec 2018

B.S. in Mathematics | Dec 2018

Relevant Coursework: Social Computing; Social Media and Data Mining; Big Data Analytics; Advanced Design and Analysis of Algorithms.

SKILLS

Research Methods: Behavioral experiments, online experiments, survey design, quantitative analysis, statistical modeling, structural equation modeling (SEM), regression, ANOVA

Programming: Python, JavaScript, R

Research Tools: Qualtrics, Prolific, Jupyter, Git

Web / Prototyping: Chrome extension development, HTML, CSS, DOM manipulation, experimental platform development

Databases / Data: MongoDB; familiarity with MySQL

Languages: Chinese (native), English (fluent)

RESEARCH EXPERIENCE

Graduate Research Assistant

Lehigh University, Bethlehem, PA | Jan 2023 – Present

Dissertation Research: From Tool to Social Actor: A Capability-Triggered Anthropomorphism Framework for General-Purpose AI

- Develops a research program on how general-purpose AI shifts from being perceived as a tool to being treated as a social actor, with a focus on perceived intelligence, anthropomorphism, and trust.
- Combines behavioral experimentation, quantitative modeling, and experimental system design to study human-AI interaction in both controlled and ecologically realistic environments.

Readable but Less Faithful: Source-Text Complexity, Trust, and Disclosure Willingness in AI Translation

- Designed and ran a 2×2 between-subjects experiment (N = 306) using TransLingo, a plain-text translation system with no avatar, voice, or persona, to test how output rendering type (readability-oriented vs. fidelity-oriented) affects perceived translation quality, with source-text complexity as a moderator of this effect, and how perceived quality is associated with perceived intelligence, anthropomorphism, trust, and self-disclosure.
- Built the TransLingo web interface, recruited participants through Prolific, ran a stimulus-level audit (Δ Flesch, Jaccard, BERTScore, COMET) to validate the rendering conditions, and analyzed results using structural equation modeling in R with lavaan.
- Found that users distinguished fidelity-oriented from readability-oriented renderings only for simple source texts, with this distinction collapsing for complex texts; perceived quality was positively associated with perceived intelligence, anthropomorphism, trust, and willingness to disclose across all six personal-information domains, consistent with an output-linked route to anthropomorphism without designed social cues.

Outer Limits: An Experimental Approach to Controlled Content Manipulation within the Reddit Interface

- Developed a Hybrid Field–Lab Methodology (HFLM) to enable controlled experiments in live social media environments while preserving stronger ecological realism than traditional lab-based studies.
- Built and validated Outer Limits, a browser-based research tool that manipulates live Reddit content and engagement cues while preserving the authentic interface for experimental use.
- Demonstrated that manipulated Reddit posts could achieve perceived realism comparable to authentic posts, supporting rigorous real-world platform experimentation.

From Tool to Social Actor: How LLM Capabilities and Social Context Shape Trust

- Designed and conducted a 2×2 between-subjects field experiment (Multifunctionality × Explanation) deployed in situ on Reddit via the Outer Limits infrastructure to test whether general-purpose AI capability cues drive users from perceiving AI as a functional tool toward perceiving it as a social actor.
- Tested a hypothesized pathway from multifunctionality and explanation to perceived intelligence and anthropomorphism, with AI literacy as a moderator and benevolence and integrity trust as socially grounded outcomes.
- Recruited active Reddit users through Prolific and analyzed the hypothesized model using structural equation modeling in R, integrating the psychological mechanism from Study 1 with the live-platform methodology from Study 2.

SELECTED PROJECTS

CellarEcho: AI-Powered Wine Assistant

- Built a personal-interest wine assistant, developed primarily through AI-assisted (agentic) coding workflows, integrating CellarTracker API data, MongoDB storage, Gemini API generation, and search-based result grounding to answer cellar-specific questions.
- Developed a chat interface with authentication, inventory synchronization, multilingual support, persistent conversation history, and real-time search calibration.
- Designed a memory layer that summarizes prior chats, extracts persistent user facts, and injects chat history, summaries, stored facts, and search signals into response generation to improve contextual continuity and answer quality.

AI Truman: Embedded AI Content and Feed Perception Study

- Contributed to the technical design of an experimental social media feed study examining how AI embeddedness and feed congruence shape attention, engagement, perceived AI prevalence, and platform trust.
- Supported research infrastructure for feed-level and post-level analyses of AI versus real content in politically congruent and incongruent online environments.

Political and Scientific Framing in Reddit Posts and Comments

- Designed and conducted a 2×2 between-subjects online field experiment (N = 442) that independently manipulated political versus scientific framing in a COVID-19 vaccination news post and its community comments, presented within the Old Reddit interface through Outer Limits.
- Examined how post and comment frames shaped self- and other-focused political and scientific frame salience, then used structural equation modeling to assess associations among frame salience, Health Belief Model perceptions, and COVID-19 vaccination intention, supplemented by coded open-ended responses.

PUBLICATIONS

- Lazard, A. J., DiFranzo, D., Nicolla, S., Aghajari, Z., Pulido, M., Mao, C., Vereen, R. N., & Dasgupta, N. (2025). Sharing antivaping social media messages. *Journal of Adolescent Health*, 77(2), 253–261. <https://doi.org/10.1016/j.jadohealth.2025.03.024>
- Nicolla, S., McKenzie, A. H., Lazard, A. J., DiFranzo, D., Aghajari, Z., Mao, C., & Dasgupta, N. (2025). Young people are willing to intervene against rape myths online: A simulated social media experiment. *Journal of Adolescent Health*. <https://doi.org/10.1016/j.jadohealth.2025.01.006>

SELECTED MANUSCRIPTS AND WORKING PAPERS

- Mao, C., Jia, H., Wegrzyn, E., & DiFranzo, D. Readable but Less Faithful: Source-Text Complexity, Trust, and Disclosure Willingness in AI Translation.
- Mao, C., Jia, H., Shi, H., Unhuryan, D. E., Baumer, E. P. S., & DiFranzo, D. Outer Limits: An Experimental Approach to Controlled Content Manipulation within the Reddit Interface.
- Zhong, W., Suk, J., Pruden, M. L., Yang, Y., Zhang, Y., Mao, C., & DiFranzo, D. Rethinking digital trace inference in political communication: Evidence from a simulated platform experiment.

TEACHING EXPERIENCE

Teaching Assistant, Computer Science Department

Lehigh University, Bethlehem, PA | Aug 2021 – Dec 2022

- Supported software engineering and introductory programming courses through office hours, grading, and student mentoring.
- Guided student teams on version control, collaboration workflows, and software project execution.
- Coordinated grading logistics for multi-section computing courses.

HONORS / CERTIFICATIONS

- Upsilon Pi Epsilon (UPE), International Honor Society for the Computing and Information Disciplines.
- Co-author on Sharing antivaping social media messages, recognized in The Distinguished Dozen: 2025 Journal of Adolescent Health Articles Making Distinguished Contributions to Adolescent and Young Adult Health.
- WSET Level 3 Award in Wines, Certified Jan 2025.